UNIT - 1

1st order LDE : $\frac{dy}{dx} + P(x)y = Q(x)$ in 'y'

$\frac{dx}{dy} + P(y)x = Q(y)$ in 'x'

$IF = e^{\int P(x)\,dx}$ (or) $e^{\int P(y)\,dy}$

Solutions : $y\,(IF) = \int Q(x)\,IF\,dx$

$x\,(IF) = \int Q(y)\,IF\,dy$

Reducible LDE : Changing independent variable

Bernoulli's Equation : $\frac{dy}{dx} + P(x)y = Q(x)y^n$ put $y^{-n} = u$

$y^{-n}\frac{dy}{dx} = a\frac{du}{dx}$ and solve the DE in 'U'

Exact DE : If DE : $M\,dx + N\,dy = 0$

(or) $\frac{dM}{dy} = \frac{dN}{dx}$ then it is exact DE

Solutions : $\int M\,dx + \int N\,dy = C$

↓ Take y terms or constant

↓ Dont write any x terms

Non Exact DE: If DE: $mdx+ndy=0$ & $\frac{dm}{dy} \neq \frac{dn}{dx}$

then DE is Non Exact

It needs to be reduced into Exact.

1. If DE is Non Exact & Homogeneous (degree of each term is same) then

   DE X IF $\Rightarrow M_1 dx + N_1 Dy = 0$ is an exact DE

   where $IF = \frac{1}{mx+ny}$

2. If DE is non exact & in the form

   $$yf(xy)dx + xg(xy)dy = 0$$

   then $IF = \frac{1}{mx-ny}$ where $mx-ny \neq 0$

3. If $mdx+ndy$ is Non exact and

i) If $\left(\frac{dm}{dy} - \frac{dn}{dx}\right)/n = f(x) \rightarrow$ Independent of y terms

then $IF = e^{\int f(x)dx}$

ii) If $\left(\frac{dm}{dy} - \frac{dn}{dx}\right)/m = g(y) \rightarrow$ Independent of x terms

then $IF = e^{\int -g(y)dy}$

Some times the Non Exact DE can be solved directly by using the derivatives of functions with x & y

Inspection method:

i, $d(xy) = x\,dy + y\,dx$

ii, $d\left(\frac{x^2+y^2}{2}\right) = x\,dx + y\,dy$

iii, $d(\log(xy)) = \frac{x\,dy + y\,dx}{xy}$

iv, $d(\log(x^2+y^2)) = \frac{x\,dx + y\,dy}{x^2+y^2}$

v, $d\left(\frac{x}{y}\right) = \frac{y\,dx - x\,dy}{y^2}$

vi, $d\left(\tan^{-1}\left(\frac{x}{y}\right)\right) = \frac{y\,dx - x\,dy}{x^2+y^2}$

Orthogonal Trajectories

Q. Given, Equation of curves with ~~[redacted]~~ constant a

Step 1: Eliminate a by differentiating on bS
If not eliminated Find a value and substitute in the curve equation. and form DE in y

Step-2: replace $\frac{dy}{dx}$ with $-\frac{dx}{dy}$

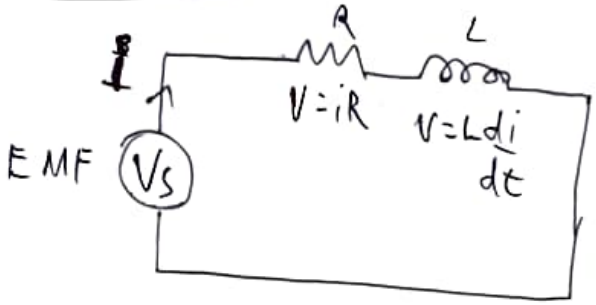
Now you got the DE of Orthogonal Trajectory of curve

Step-3: Solve the DE of OT

If the DE in y and DE of OT are same then the curve is Self orthogonal

If the DE is in $\frac{dr}{d\theta}$ then replace it with $-r^2\frac{d\theta}{dr}$

Electrical Circuits



By applying KVL

$$L\frac{di}{dt} + Ri = EMF$$

$$\frac{di}{dt} + \frac{R}{L}i = \frac{E}{L}$$

is the DE of LR circuit

Independent variable = t Dependent variable = i

$$\Rightarrow \boxed{\frac{di}{dt} + P(t)i = Q(t)}$$

UNIT-2

Higher order LDE:

$$\frac{d^n y}{dx^n} + P_1 \frac{d^{n-1} y}{dx^{n-1}} + P_2 \frac{d^{n-2} y}{dx^{n-2}} + \cdots P_n y = Q(x)$$

Solution: $y = CF + PI$

Finding CF: Let $\frac{d^n y}{dx^n} = D^n$

Operator form: $f(D)\, y = Q(x)$

Auxilary equation: $f(m) = 0$

Find roots

1. If roots are real & Distinct, $m_1 \neq m_2 \neq m_3 \cdots$

$$CF = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x} + \cdots$$

2. If roots are equal, $m_1 = m_2 = m_3 \cdots$

$$CF = (C_1 + C_2 x + C_3 x^2 + \cdots) e^{mx}$$

3. If roots are $a+ib$, $a-ib$

$$CF = (C_1 \cos bx + C_2 \sin bx) e^{ax}$$

4. If roots are $a \pm ib$, $c \pm id$

$$CF = [(C_1 + C_2 x)\cos bx + (C_1 + C_2 x)\sin bx]\, e^{ax} + [(C_1 + C_2 x)\cos dx + (C_1 + C_2 x)\sin dx]\, e^{cx}$$

Finding PI

$$PI = \frac{1}{f(D)} Q(x)$$

i) $Q(x) = e^{ax}$ put $D = a$

If $f(a) = 0$ then $PI = \frac{x}{f'(a)} Q(x)$

If $f'(a) = 0$ then $PI = \frac{x^2}{f''(a)} Q(x)$ & soon

ii) If $Q(x) = \sin ax / \cos ax$ functions put $D^2 = -a^2$
only evaluate if there is D^2

iii) If $Q(x) = x^m$ then

$$PI = \frac{1}{f(D)} x^m = \frac{1}{[1+\phi(D)]^n} x^m$$

$$= [1+\phi(D)]^{-n} x^m$$

$(1+x)^{-1} = [1 - x + x^2 - x^3 + \cdots]$ until x^m the term

$(1-x)^{-1} = [1 + x + x^2 + x^3 + \cdots]$

$(1-x)^{-2} = [1 + 2x + 3x^2 + 4x^3 \cdots]$

$(1+x)^{-2} = [1 - 2x + 3x^2 - 4x^3 + \cdots]$

iv) If $Q(x) = e^{ax} V(x)$

Then $PI = e^{ax} \times \frac{1}{f(D+a)} \times V(x)$

v) If $Q(x) = x(V(x))$

Then $PI = \left[x - \frac{f'(D)}{f(D)}\right] \frac{V(x)}{f(D)}$

Variation of Parameters

DE: $F(D)y = Q(x) = R$

then $CF = C_1(u(x)) + C_2(v(x))$

Wronstein $W = \begin{vmatrix} U & V \\ U' & V' \end{vmatrix}$

$PI = AU + BV$ where $A = -\int \frac{VR}{W}$ $B = \int \frac{UR}{W}$

Solution: $y = CF + PI$